

Lab 6

Weiwei Wang

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Deadline: 2022-11-13 22:30:00 Beijing Time

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Exercise06_01.java

(Math: pentagonal numbers) A pentagonal number is defined as $n(3n - 1)/2$ for $n = 1, 2, \dots$, and so on. Write a method with the following header that returns a pentagonal number:

```
public static int getPentagonalNumber(int n)
```

Write a test program that uses this method to display the first 100 pentagonal numbers with 10 numbers on each line. Use the %7d format to display each number.

```
    1      5      12      22      35      51      70      92      117      145
  176    210    247    287    330    376    425    477    532    590
...
12376 12650 12927 13207 13490 13776 14065 14357 14652 14950
```

Exercise06_02.java

(Sum the digits in an integer) Write a method that computes the sum of the digits in an integer. Use the following method header:

```
public static int sumDigits(long n)
```

For example, `sumDigits(234)` returns 9 (= 2 + 3 + 4). (Hint: Use the % operator to extract digits and the / operator to remove the extracted digit. Write a test program that prompts the user to enter an integer then displays the sum of all its digits.

```
Enter a number: 1234  
The sum of digits for 1234 is 10
```

Exercise06_03.java

(Palindrome integer) Write the methods with the following headers:

```
// Return the reversal of an integer, e.g., reverse(456) returns 654
public static int reverse(int number)
// Return true if number is a palindrome
public static boolean isPalindrome(int number)
```

Use the reverse method to implement isPalindrome. A number is a palindrome if its reversal is the same as itself. Write a test program that prompts the user to enter an integer and reports whether the integer is a palindrome.

Exercise06_13.java

(Sum series) Write a method to compute the following series

$$m(i) = \frac{1}{2} + \frac{2}{3} + \dots + \frac{i}{i+1}$$

Write a method with the following header

```
public static double m(int i)
```

Write a test program that displays the following table

i	m(i)
1	0.5000
2	1.1667
...	
20	17.3546

Exercise06_36.java

(Geometry: area of a regular polygon) A regular polygon is an n -sided polygon in which all sides are of the same length and all angles have the same degree. The formula for computing the area of a regular polygon is

$$\text{Area} = \frac{n \times s^2}{4 \tan(\pi/n)}$$

Write a method that returns the area of a regular polygon using the following header:

```
public static double area(int n, double side)
```

Write a main method that prompts the user to enter the number of sides and the side of a regular polygon and displays its area. Here is a sample run:

```
Enter the number of sides: 5 [Enter]
Enter the side: 6.5 [Enter]
The area of the polygon is 72.69017017488385
```

Exercise07_03.java

(Count occurrence of numbers) Write a program that reads integers and counts the occurrences of the numbers between 1 and 50. Assume the input ends with 0. Here is a sample run of the program:

```
Enter the integers between 1 and 50: 1 2 100 4 0 [Enter]
1 occurs 1 time
2 occurs 1 time
4 occurs 1 time
```

Exercise07_08.java

(Average an array) Write a method that return the average of an array with the following header:

```
public static double average(double[] array)
```

Write a test program that prompts the user to enter 10 double values, invokes this method, then displays the average value.

Here is the sample run

```
Enter 10 double values: 1 2 3 4 5 6 7 8 9 10  
5.5
```

Exercise07_09.java

(Find the largest element) Write a method that finds the largest element in an array of double values using the following header:

```
public static double max(double[] array)
```

Write a test program that prompts the user to enter ten numbers, invokes this method to return the maximum value, and displays the maximum value. Here is a sample run of the program:

```
Enter ten numbers: 1 2 100 1 2 3 4 5 -100 9  
The maximum number is 100.0
```